

Construction and validation of a profile hidden Markov model for the detection of the Kunitz domain

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1 Introduction

Sequence database homology searching is one of the most powerful tools in computational molecular biology, enabling the detection of remote evolutionary relationships between protein sequences. Among the probabilistic approaches developed for this purpose, Profile Hidden Markov Models (HMMs) have proven particularly effective for protein domain recognition. Unlike pairwise alignment tools such as BLAST, which rely on a context-independent substitution matrix, a profile HMM encodes position-specific amino acid frequencies and gap penalties derived from a multiple sequence alignment (MSA), capturing the evolutionary constraints characteristic of a specific protein family [1]. This position-specific scoring makes profile HMMs especially suited for detecting divergent or remote homologs that would be missed by simpler similarity searches.

The target of this work is the Kunitz domain (PFAM: PF00014), one of the most extensively studied protein domains in structural biology. The domain was first identified by [2] in the bovine pancreatic trypsin inhibitor (BPTI), isolated from beef pancreas. Kunitz-domain inhibitors are found in animals, plants, and microbes [3], and are typically 50–70 amino acids in length. They adopt a conserved structural fold consisting of two antiparallel β -sheets and one or two helical regions, stabilized by three disulfide bridges with the bonding pattern C1–C6, C2–C4, C3–C5 [4]. These disulfide bridges maintain the structural integrity of the inhibitor and present the protease-binding loop at the protein surface. A highly exposed P1 residue at position 15 (typically Arg or Lys) inserts into the S1 pocket of the target protease and is the primary determinant of inhibitory specificity [5].

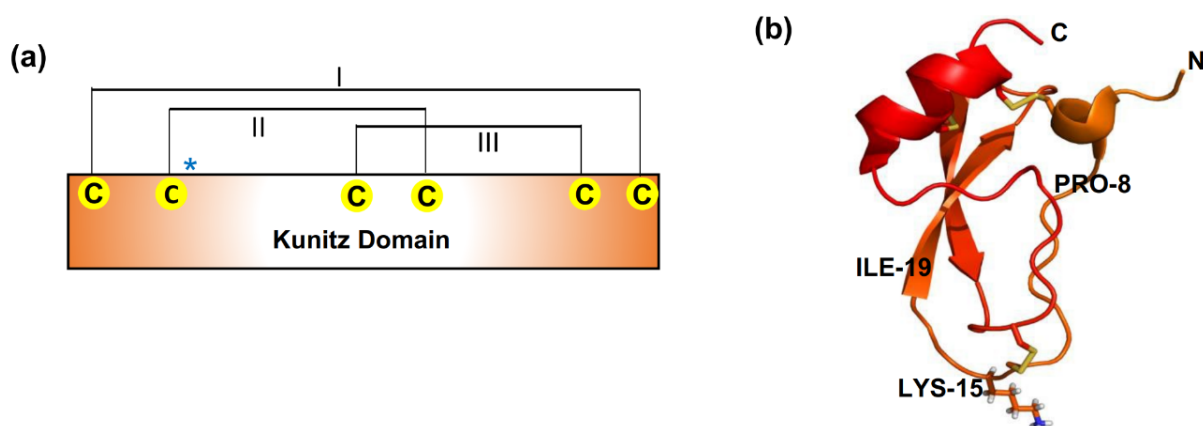


Figure 1: Structure and activity of Kunitz-domain inhibitors. (a) Schematic representation of Kunitz-domain inhibitor showing the disulfide bonding pattern of 1-6, 2-4, 3-5. Conserved cysteine residues are marked with yellow and the P1 active site residue, the primary determinant of the specificity of serine protease inhibition is marked with a blue asterisk. (b) Crystal structure of BPTI (PDB: 4Y0Y) showing three disulfide bonds (in yellow), the solvent-exposed loop (P8 to I19) highly complementary to the enzyme active site, and the P1 residue (K15).

2 Materials and Methods

2.1 Dataset Construction

Experimentally determined protein structures containing the Kunitz domain were retrieved from the RCSB Protein Data Bank (RCSB PDB) [6] using the following search criteria: Annotation Identifier = “PF00014”, Annotation Type = “Pfam”, data collection resolution ≤ 3.5 Å, and polymer entity sequence length between 40 and 70 amino acids. Results were downloaded in tabular format including the following fields: Entry ID, Sequence, Auth Asym ID, Entity ID, and Annotation Identifier. The initial query returned 134 PDB entries, corresponding to 6,229 rows in the tabular file (multiple rows per entry due to multiple annotation records per chain).

The tabular output was processed with a custom Python script. First, only rows with Annotation Identifier equal to “PF00014” were retained, in order to exclude co-crystallized chains that satisfied the resolution and length filters but did not carry the Kunitz domain annotation. This step was necessary because the PDB query returns all polymer entities of a structure when at least one entity satisfies the search criteria, including non-Kunitz co-crystallized partners. A secondary sequence length filter (40–80 amino acids) was retained as a conservative criterion to exclude potential domain fragments (sequences shorter than 40 aa) and chains with extensive flanking regions (sequences longer than 80 aa). Three sequences were excluded by this filter: 5JBT:Y (38 aa, covering only positions 323–360 of a Kunitz domain annotated at positions 306–364 in UniProt Q06481, suggesting an incomplete fragment), and 5NX1:C and 5NX3:C (81 aa each, both mapping to the Kunitz domain of APP, UniProt P05067, positions 289–346). The latter two sequences originate from the same protein as the training set structure 1AAP:A (also APP, UniProt P05067), which has a superior crystallographic resolution of 1.50 Å compared to 5NX1 (1.85 Å) and 5NX3 (2.30 Å). Consequently, 1AAP:A would have been selected as cluster representative regardless of whether 5NX1:C and 5NX3:C were included, and their exclusion did not alter the final training set composition. After filtering, 133 unique Kunitz domain chains were retained.

2.2 Redundancy Removal

The 133 chains were converted to FASTA format using a custom `awk` script. Sequence clustering was performed with MMseqs2 [7] using a minimum sequence identity threshold of 0.8 and a minimum alignment coverage of 0.8. These parameters were chosen for profile HMM construction, as a trade-off between redundancy reduction and preservation of sequence diversity. Clustering yielded 12 clusters.

The default MMseqs2 representative selection maximizes the number of sequences covered per cluster without considering crystallographic quality. Since the representative structures are subsequently used to build a multiple structure alignment, we implemented a custom Python script that selects, for each cluster, the structure with the lowest crystallographic resolution value (i.e., highest structural quality), querying resolution data via the RCSB PDB GraphQL API (`rscsb_entry_info.resolution_combined` field). In the case of ties, the sequence with the greatest length was selected as tiebreaker. This procedure yielded 12 representative structures.

2.3 Multiple Structure Alignment and Iterative Refinement

The 12 representative PDB structures were downloaded from RCSB PDB and the chains of interest were extracted using a custom Python script, removing all co-crystallized molecules not corresponding to the Kunitz domain chain. Multiple structure alignment was performed with mTM-align [8].

An iterative refinement procedure was applied to improve the quality of the alignment, following two complementary criteria:

RMSD-based criterion. For each structure, the mean pairwise RMSD with respect to all other structures was computed from the mTM-align output matrix. Structures with a mean RMSD exceeding the global mean plus one standard deviation were considered outliers and removed. This conservative threshold (mean + 1 SD) was preferred over the simpler mean threshold to avoid over-removal of structures from an already small dataset.

Cysteine alignment criterion. The six conserved cysteine residues forming the three characteristic disulfide bridges of the Kunitz domain (pattern C1–C6, C2–C4, C3–C5) were used as structural anchors to evaluate alignment quality. Sequences with cysteine residues consistently misaligned with respect to the consensus positions were flagged for removal.

Additionally, manual inspection of the multiple sequence alignment derived from the multiple structure alignment revealed two pairs of nearly identical sequences originating from the same protein: 3FP7_J and 1G6X_A (both Pancreatic trypsin inhibitor, UniProt P00974), and 5NX1_D and 1AAP_A (both Amyloid precursor protein, UniProt P05067). These pairs had not been merged by MMseqs2 because the shorter sequences (3FP7_J and 5NX1_D) exhibited an alignment coverage below the 0.8 threshold with respect to their longer counterparts. Redundancy was resolved manually by retaining the representative with the lowest crystallographic resolution and complete sequence coverage of all six cysteine residues, leading to the removal of 3FP7_J and 5NX1_D.

One sequence, 3WNY_A, displayed an atypical cysteine pattern (only two cysteines aligned to positions C1 and C6 of the canonical motif). Despite this divergence, 3WNY_A was retained because it carries a validated PF00014 annotation in UniProt, and its inclusion enriches the model with information about tolerated positional variability. Structure 4BQD_A presented an N-terminal linker region (preceding the Kunitz domain as annotated in UniProt P10646) that did not align with any other sequence. This region was not removed manually, as it represents less than 50% of any aligned column and is therefore treated as an insert state by `hmmbuild`, contributing no match states to the final model.

After the iterative refinement process, the final training set comprised 10 structures. The final multiple structure alignment yielded the following quality metrics: $L_{\text{core}} = 51$, ccRMSD = 0.70 Å, ccTM-score = 0.85, $L_{\text{ali}} = 55.04$, RMSD = 0.84 Å, TM-score = 0.90.

2.4 HMM Model Construction

The profile HMM was built from the multiple sequence alignment derived from the final multiple structure alignment using `hmmbuild` from the HMMER3 package [1] with default parameters. In the resulting model, alignment columns in which more than 50% of sequences carry a gap are treated as insert states and do not contribute emission probabilities to the profile; the remaining columns become match states encoding position-specific amino acid frequencies.

2.5 Validation Dataset

Positive and negative sequence datasets were retrieved from the UniProt/Swiss-Prot database [UniProt2023] (reviewed entries only), ensuring high annotation reliability. The positive set was defined by the query `(reviewed:true) AND (xref:pfam-PF00014)`, yielding sequences annotated with the Kunitz domain. The negative set was defined by `(reviewed:true) NOT (xref:pfam-PF00014)`, comprising all reviewed sequences lacking this annotation.

To avoid data leakage, the sequences corresponding to the 10 structures used for model training were removed from the positive set. PDB chain identifiers were mapped to UniProt accession numbers using the UniProt ID Mapping tool, and the corresponding entries were excluded from the positive set using a custom Python script; removal was verified by confirming the absence of any overlap between training and positive set identifiers using the `comm` Unix utility. The final positive set comprised 390 sequences. The negative set comprised 574,229 sequences, resulting

in a heavily imbalanced dataset; the Matthews Correlation Coefficient (MCC) was therefore adopted as the primary performance metric, as it is robust to class imbalance unlike accuracy.

2.6 Cross-Validation and Threshold Optimization

Homology searches were performed with `hmmsearch` from HMMER3 [1] using the `-max` option to disable heuristic acceleration filters and maximize sensitivity, and `-Z 1000` to normalize the database size for comparable E-value computation across both datasets. The best domain E-value (column 8 of the `hmmsearch` tabular output) was used as the classification score in preference to the full-sequence E-value, as the Kunitz domain frequently occurs as an individual domain embedded within larger multidomain proteins; the domain-level score is therefore more informative for domain detection.

Sequences in the negative set that produced no match were assigned an E-value of 100 (substantially above the maximum observed E-value of 6.9) and labelled as true negatives. Both datasets were randomly shuffled prior to cross-validation (`sort -R`).

A two-fold cross-validation was performed by partitioning both the positive and negative sets into two equal folds, each containing 195 positive sequences and 287,115 negative sequences. In each iteration, one fold was used to optimize the E-value classification threshold and the complementary fold served as the validation set. Threshold optimization was performed by evaluating MCC over a grid of E-value thresholds ranging from 10^{-1} to 10^{-15} . The optimal threshold was selected as the value yielding the highest MCC across both folds. A threshold of 10^{-5} was identified as optimal, producing consistent MCC values of 0.9923 on both folds.

3 Results

3.1 Training Set Composition

Starting from 134 PDB entries retrieved by the initial query, the filtering pipeline retained 133 unique Kunitz domain chains. After clustering with MMseqs2 at 80% sequence identity and 80% coverage, 12 representative structures were selected based on crystallographic resolution. Table 1 lists the 10 structures constituting the final training set after the iterative MSA refinement process, together with their PDB identifiers, chain identifiers, source organisms, and crystallographic resolutions.

Table 1: Final training set of 10 Kunitz domain structures.

PDB ID	Chain	Organism	Resolution (Å)
1AAP	A	<i>Homo sapiens</i>	1.50
1G6X	A	<i>Bos taurus</i>	0.86
1KTH	A	<i>Homo sapiens</i>	0.95
1ZR0	B	<i>Homo sapiens</i>	1.80
3WNY	A	<i>Bos taurus</i>	1.30
4BQD	A	<i>Homo sapiens</i>	2.48
4DTG	K	<i>Homo sapiens</i>	1.80
4ISO	B	<i>Homo sapiens</i>	2.10
4U30	X	<i>Homo sapiens</i>	2.50
4U32	X	<i>Homo sapiens</i>	1.65

The iterative refinement of the multiple structure alignment led to the removal of two sequences removed for manual redundancy resolution (3FP7_J and 5NX1_D). The final alignment of 10 structures achieved a TM-score of 0.90 and an RMSD of 0.84 Å (Table 2), reflecting the high structural conservation characteristic of the Kunitz domain fold.

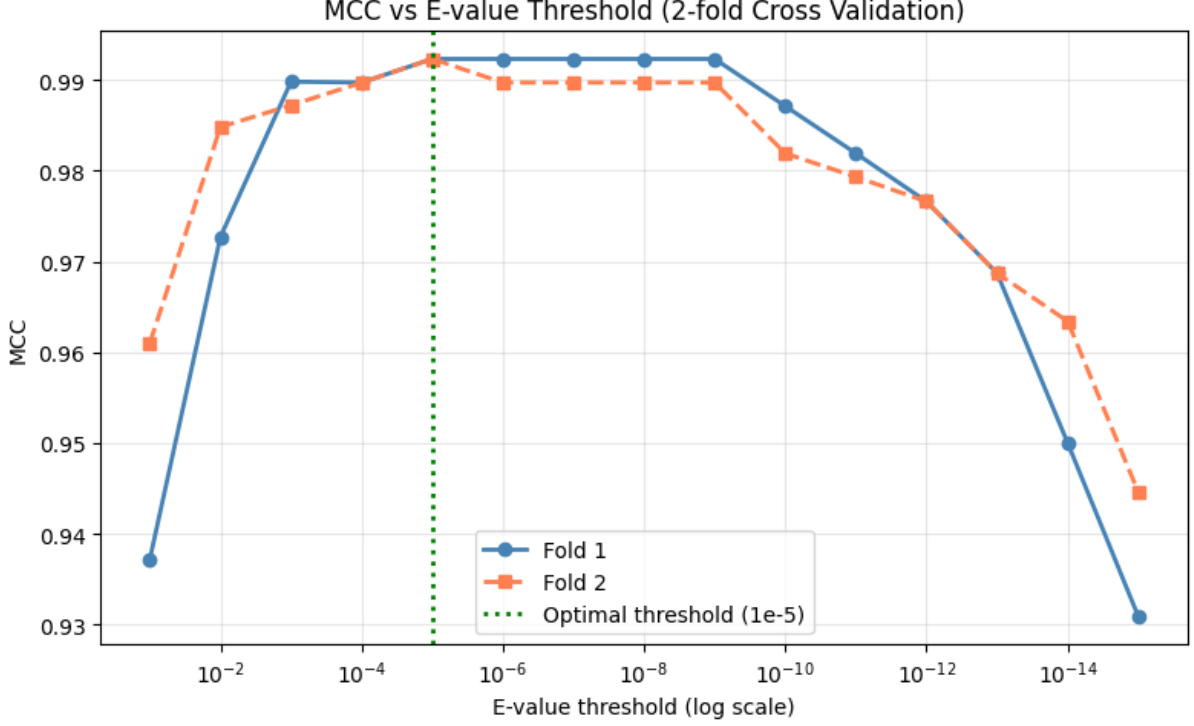


Figure 2

Table 2: Quality metrics of the final multiple structure alignment (mTM-align [8]).

Metric	Value
L_{core}	51
ccRMSD	0.70 Å
ccTM-score	0.85
L_{ali}	55.04
RMSD	0.84 Å
TM-score	0.90

3.2 Model Performance

The profile HMM was evaluated on both dataset folds (195 positive sequences and 287,115 or 287,114 negative sequences) using the optimal E-value threshold of 10^{-5} identified by two-fold cross-validation. Figure 2 shows the MCC as a function of the E-value threshold for both cross-validation folds, demonstrating consistent performance and a clear optimum at 10^{-5} .

The confusion matrices for the two validation folds are reported in Figure 3. The ROC curves for both folds are shown in Figure 4, yielding AUC values close to 1.0, confirming the high discriminative power of the model across all possible threshold values.

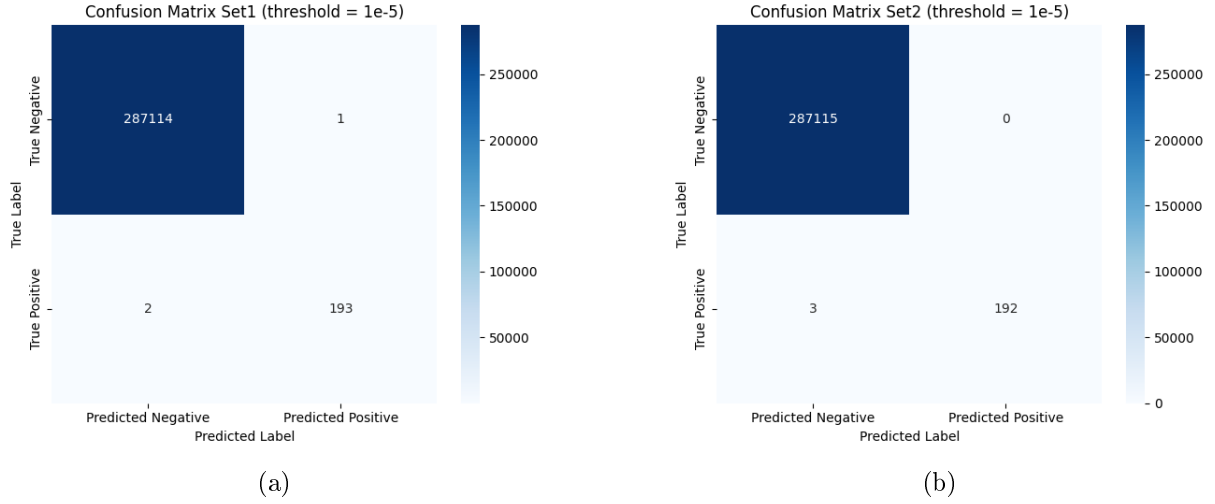


Figure 3: Confusion matrices of both set1 (a) and set2 (b).

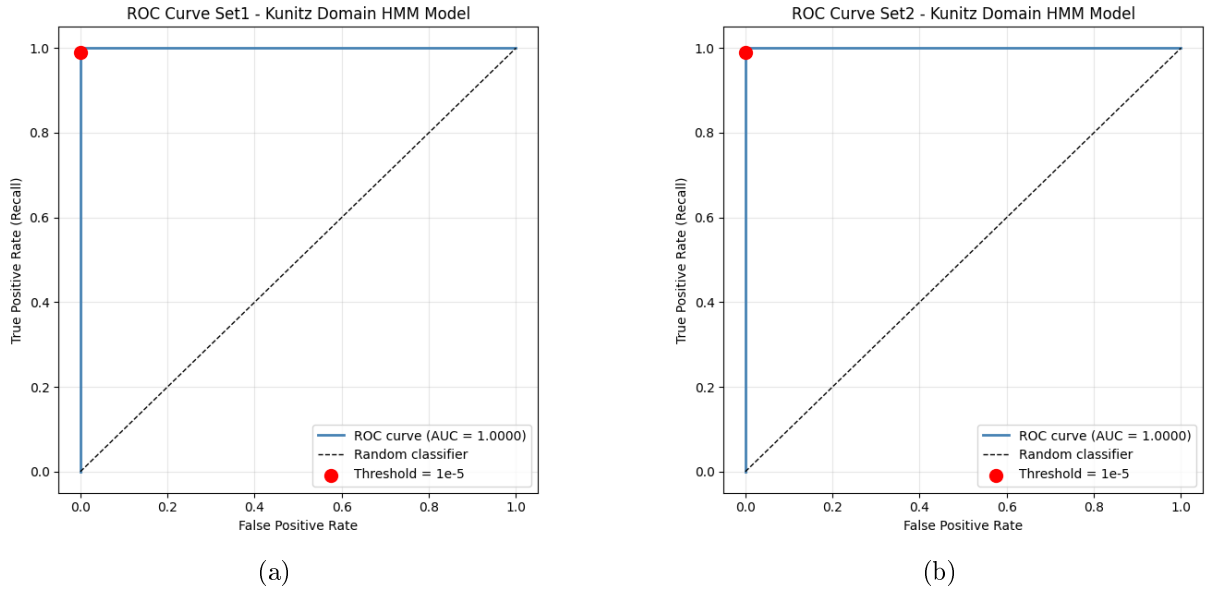


Figure 4: ROC curves of both set1 (a) and set2 (b).

3.3 Analysis of False Negatives

Five sequences annotated as Kunitz domain in UniProt were not detected by the model at the optimal threshold (Table 3). All five carry computational annotations (PROSITE-ProRule PRU00031) rather than direct experimental evidence, introducing some uncertainty about their true classification status.

Table 3: False negative sequences with best domain E-value and annotation evidence.

UniProt ID	Protein	E-value	# Kunitz	Annotation
Q8MVC4	Penthalaris	1.4×10^{-5}	5	PROSITE-ProRule
Q8WPG5	Savignin	2.1×10^{-4}	2	PROSITE-ProRule
A0A1Q1NL17	HA11	8.7×10^{-3}	1	PROSITE-ProRule
O62247	BLI-5 (<i>C. elegans</i>)	1.2×10^{-2}	1	PROSITE-ProRule
D3GGZ8	BLI-5 (<i>H. contortus</i>)	1.4×10^{-1}	1	PROSITE-ProRule

Structural analysis using AlphaFold-predicted structures [9] confirmed that D3GGZ8 and O62247 are structurally divergent from the training set (pairwise RMSD of 1.55–1.85 Å versus the training set average), yet highly similar to each other (RMSD = 0.35 Å). Both sequences originate from nematodes (*Haemonchus contortus* and *Caenorhabditis elegans*), organisms evolutionarily distant from the vertebrates dominating the training set. Furthermore, both lack cysteines at positions C2 and C4, losing the central disulfide bridge (C2–C4) of the canonical pattern. The combination of high structural divergence, atypical cysteine pattern, and evolutionary distance suggests these sequences represent remote homologs of the Kunitz domain that are not well represented in the training set. An additional contributing factor is the position of the Kunitz domain within the full-length protein sequence: both D3GGZ8 and O62247 carry the domain at positions 120–190 and 135–184, respectively, within proteins of approximately 195–202 amino acids. The presence of flanking non-Kunitz sequence increases the effective length used for E-value computation, penalizing the domain-level score.

Q8WPG5 shows high structural divergence from the training set (RMSD = 2.1–2.7 Å) despite a very high AlphaFold pLDDT score in the Kunitz domain region, suggesting a genuinely atypical fold rather than a prediction artifact. It lacks cysteine C4, disrupting the C2–C4 disulfide bridge.

Q8MVC4 represents a borderline case: with five Kunitz domains and a best domain E-value of 1.4×10^{-5} , it falls just above the optimal threshold of 10^{-5} . A marginally more permissive threshold would correctly classify this sequence.

A0A1Q1NL17 shows intermediate divergence (RMSD = 1.2–1.5 Å) and lacks cysteine C2. Its relatively high E-value (8.7×10^{-3}) likely reflects sequence divergence from the training set rather than a structural anomaly.

The single false positive was not investigated in detail given the very low false positive rate (1 out of 574,229 negative sequences).

4 Discussion and Conclusions

In this work, a profile HMM for the detection of the Kunitz domain (PF00014) was constructed from a set of 10 experimentally determined structures and evaluated on a large, curated sequence dataset from UniProt/Swiss-Prot. The model achieves an MCC of 0.9923, demonstrating high discriminative power despite the severe class imbalance between positive (390) and negative (574,229) sequences.

Several methodological choices contributed to the quality of the final model. First, the use of crystallographic resolution as the criterion for selecting cluster representatives, rather than the default approach of MMseqs2, ensured that the training structures provided the most reliable atomic coordinates for multiple structure alignment. Second, the combination of RMSD-based and cysteine-alignment criteria in the iterative refinement of the MSA allowed the removal of structurally inconsistent sequences while retaining legitimate variants such as 3WNY_A, which displays an atypical cysteine pattern yet carries a validated Kunitz domain annotation. Third, the use of the best domain E-value rather than the full-sequence E-value for classification correctly accounts for the frequent occurrence of the Kunitz domain as an embedded domain within larger multidomain proteins.

An important observation emerging from the false negative analysis is the divergence between the RMSD-based outlier criterion and the cysteine alignment criterion during MSA refinement. These two criteria identified different candidate structures for removal, highlighting that structural divergence at the global level (RMSD) does not necessarily correspond to disruption of the functionally critical cysteine positions, and vice versa. This observation suggests that a combined evaluation is necessary for robust training set curation.

The five false negatives all carry computational annotations (PROSITE-ProRule PRU00031) rather than direct experimental evidence, and four of them display structural or sequence features suggesting divergence from canonical Kunitz domain members: atypical disulfide bridge

patterns, evolutionary distance from the training set organisms, or domain positioning within long multidomain proteins. These findings suggest that the false negatives may represent remote homologs or divergent variants that are underrepresented in the training set, which was predominantly built from vertebrate structures.

Several limitations should be acknowledged. The training set comprises only 10 structures, predominantly from vertebrates, which may limit the model’s sensitivity towards evolutionarily divergent Kunitz domain sequences from other taxa. Additionally, the clustering parameters (identity = coverage = 0.8) were fixed; exploring a range of values (0.8–0.95) could yield models with different sensitivity-specificity trade-offs. Future work should address these limitations by expanding the training set with structurally characterized Kunitz domains from non-vertebrate organisms and by performing a systematic evaluation across multiple clustering parameter settings.

In conclusion, the profile HMM constructed in this work accurately detects the Kunitz domain in Swiss-Prot reviewed sequences, with very few misclassifications. The methodological pipeline, from structure-guided training set curation to cross-validated threshold optimization, provides a reproducible framework applicable to the construction of profile HMMs for other protein domain families.

References

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A Additional materials

All the materials about this project can be found in this github repository: <https://github.com/Valentina-Piccoli/Kunitz-domain-project.git>

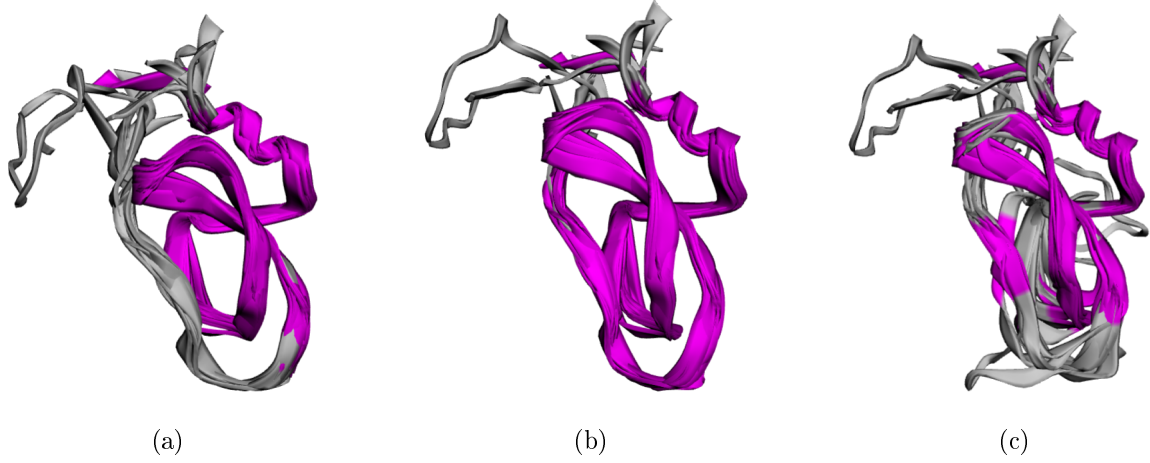


Figure 5: **3D representation of the multiple structures alignment with mTM-align2.** (a) represents the MSA obtained with all 12 structures remained after the clustering procedure, while (b) represents the one made with the best 10 structures selected for train the hidden markov model. (c) represents the structural alignment of the 10 structures used to build the model and 4 false negatives (the one investigated to explain the misclassification); it can be seen that the core region highlighted in purple is clearly reduced when we introduce also the false negative, meaning that they diverge in structures with respect to the one represented by our model.

	1AAP_A.pdb	1G6X_A.pdb	1KTH_A.pdb	1ZR0_B.pdb	3FP7_J.pdb	3WNY_A.pdb	4BQD_A.pdb	4DTG_K.pdb	4ISO_B.pdb	4U30_X.pdb	4U32_X.pdb	5NX1_D.pdb
1AAP_A.pdb	0.000	0.971	1.162	0.717	0.666	1.048	0.723	0.787	0.957	0.969	0.503	0.322
1G6X_A.pdb	0.971	0.000	1.305	1.217	0.636	0.458	1.184	1.034	1.361	0.977	0.601	0.734
1KTH_A.pdb	1.162	1.305	0.000	1.365	1.326	1.004	1.254	1.437	1.655	1.182	0.875	0.874
1ZR0_B.pdb	0.717	1.217	1.365	0.000	1.221	1.109	1.271	1.388	1.426	0.838	0.764	0.720
3FP7_J.pdb	0.666	0.636	1.326	1.221	0.000	0.556	1.046	1.038	1.281	0.982	0.621	0.826
3WNY_A.pdb	1.048	0.458	1.004	1.109	0.556	0.000	0.788	0.876	1.216	1.199	0.734	0.829
4BQD_A.pdb	0.723	1.184	1.254	1.271	1.046	0.788	0.000	1.060	1.272	0.958	0.488	0.657
4DTG_K.pdb	0.787	1.034	1.437	1.388	1.038	0.876	1.060	0.000	1.589	0.927	0.548	0.615
4ISO_B.pdb	0.957	1.361	1.655	1.426	1.281	1.216	1.272	1.589	0.000	1.012	0.452	0.709
4U30_X.pdb	0.969	0.977	1.182	0.838	0.982	1.199	0.958	0.927	1.012	0.000	1.048	0.838
4U32_X.pdb	0.503	0.601	0.875	0.764	0.621	0.734	0.488	0.548	0.452	1.048	0.000	0.416
5NX1_D.pdb	0.322	0.734	0.874	0.720	0.826	0.829	0.657	0.615	0.709	0.838	0.416	0.000

Figure 6: **Table with pairwise RMSD scores for the first MSA** (the one with all 12 structures obtained after the clustering). I used a python script to evaluate which structures has an average RMSD higher than the global average RMSD plus one standard deviation, and I obtain as result the two structures 1KTH:A and 4ISO:B (with respectively 1.221727 and 1.175455 as average RMSD). Both 1KTH:A and 4ISO:B have an higher average RMSD but looking to the RMSD table we can see that the higher pairwise RMSD values are with specific structures, not with all. This means that their 3D structure diverges just with soem particular structures and not globally. For this reason I decide to keep them in the set of structures used to build the hidden markov model.

	1AAP_A.pdb	1G6X_A.pdb	1KTH_A.pdb	1ZR0_B.pdb	3WNY_A.pdb	4BQD_A.pdb	4DTG_K.pdb	4ISO_B.pdb	4U30_X.pdb	4U32_X.pdb
1AAP_A.pdb	0.000	0.971	1.162	0.717	1.048	0.723	0.787	0.957	0.969	0.503
1G6X_A.pdb	0.971	0.000	1.305	1.217	0.458	1.184	1.034	1.361	0.977	0.601
1KTH_A.pdb	1.162	1.305	0.000	1.365	1.004	1.254	1.437	1.655	1.182	0.875
1ZR0_B.pdb	0.717	1.217	1.365	0.000	1.109	1.271	1.388	1.426	0.838	0.764
3WNY_A.pdb	1.048	0.458	1.004	1.109	0.000	0.788	0.876	1.216	1.199	0.734
4BQD_A.pdb	0.723	1.184	1.254	1.271	0.788	0.000	1.060	1.272	0.958	0.488
4DTG_K.pdb	0.787	1.034	1.437	1.388	0.876	1.060	0.000	1.589	0.927	0.548
4ISO_B.pdb	0.957	1.361	1.655	1.426	1.216	1.272	1.589	0.000	1.012	0.452
4U30_X.pdb	0.969	0.977	1.182	0.838	1.199	0.958	0.927	1.012	0.000	1.048
4U32_X.pdb	0.503	0.601	0.875	0.764	0.734	0.488	0.548	0.452	1.048	0.000

Figure 7: **Table with pairwise RMSD scores for the second MSA** (the one with just the 10 best strcutres used to build the hidden markov model). By using the same python script that evaluates which structures has an average RMSD higher than the global average RMSD plus one standard deviation, i obtain as result the same two structures as before (1KTH:A and 4ISO:B). For the same reasoning explained in the caption of the previous image, I decide to keep them.

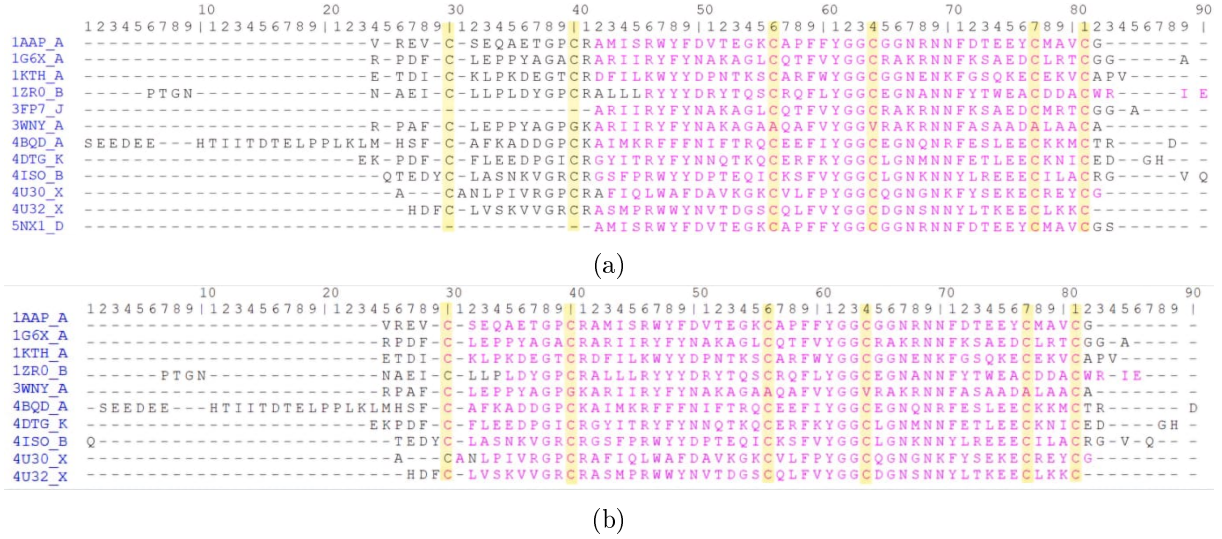


Figure 8: **Multiple sequence alignment obtained from multiple structure alignment with mTM-align2**. The first alignment is made with all 12 structures remained after the clustering procedure, while the second alignment show only the sequences of the best 10 structures selected for train the hidden markov model. The pink region shows the common core (most similar region with lowest RMSD across structures). The columns highlighted in yellow are the one corresponding to the six cysteines.

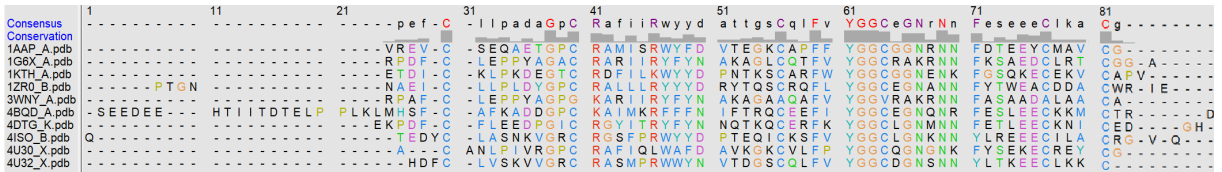
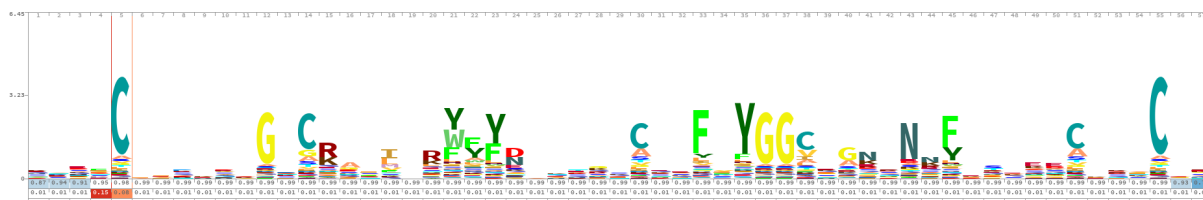
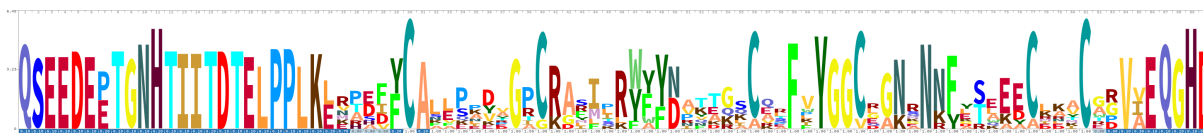


Figure 9: **Different representation of the multiple sequence alignment obtained from multiple structure alignment**. The conservation level in each position of the alignment is represented as grey bargraph.



	1AAP_A.pdb	1G6X_A.pdb	1KTH_A.pdb	1ZR0_B.pdb	3WNY_A.pdb	4BQD_A.pdb	4DTG_K.pdb	4ISO_S.pdb	4U30_X.pdb	4U32_X.pdb	A0A1Q1NL17_kunitz.pdb	D3GGZ8_kunitz.pdb	O62247_kunitz.pdb	Q8WP65_kunitz.pdb
1AAP_A.pdb	0.000	0.971	1.162	0.717	1.048	0.723	0.787	0.957	0.969	0.503	1.487	1.550	1.591	2.501
1G6X_A.pdb	0.971	0.000	1.305	1.217	0.458	1.184	1.034	1.361	0.977	0.601	1.487	1.844	1.651	2.420
1KTH_A.pdb	1.162	1.305	0.000	1.365	1.004	1.254	1.437	1.655	1.182	0.875	1.219	1.849	1.456	2.637
1ZR0_B.pdb	0.717	1.217	1.365	0.000	1.109	1.271	1.388	1.426	0.838	0.764	1.499	1.854	1.588	2.683
3WNY_A.pdb	1.048	0.458	1.004	1.109	0.000	0.788	0.876	1.216	1.199	0.734	1.438	1.863	1.690	2.408
4BQD_A.pdb	0.723	1.184	1.254	1.271	0.788	0.000	1.060	1.272	0.958	0.488	1.473	2.006	1.616	2.142
4DTG_K.pdb	0.787	1.034	1.437	1.388	0.876	1.060	0.000	1.589	0.927	0.548	1.414	1.969	1.622	2.178
4ISO_S.pdb	0.957	1.361	1.655	1.426	1.216	1.272	1.589	0.000	1.012	0.452	1.251	1.747	1.443	2.516
4U30_X.pdb	0.969	0.977	1.182	0.838	1.199	0.958	0.927	1.012	0.000	1.048	1.437	1.813	1.508	2.429
4U32_X.pdb	0.503	0.601	0.875	0.764	0.734	0.488	0.548	0.452	1.048	0.000	1.377	1.613	1.518	2.090
A0A1Q1NL17_kunitz.pdb	1.487	1.357	1.219	1.499	1.438	1.473	1.514	1.251	1.437	1.377	0.000	1.448	1.702	2.405
D3GGZ8_kunitz.pdb	1.550	1.844	1.849	1.854	1.863	2.006	1.969	1.747	1.813	1.613	1.448	0.000	0.353	2.890
O62247_kunitz.pdb	1.591	1.651	1.456	1.588	1.690	1.616	1.622	1.443	1.508	1.518	1.702	0.353	0.000	2.609
Q8WP65_kunitz.pdb	2.501	2.420	2.637	2.683	2.408	2.142	2.178	2.516	2.429	2.090	2.405	2.890	2.609	0.000

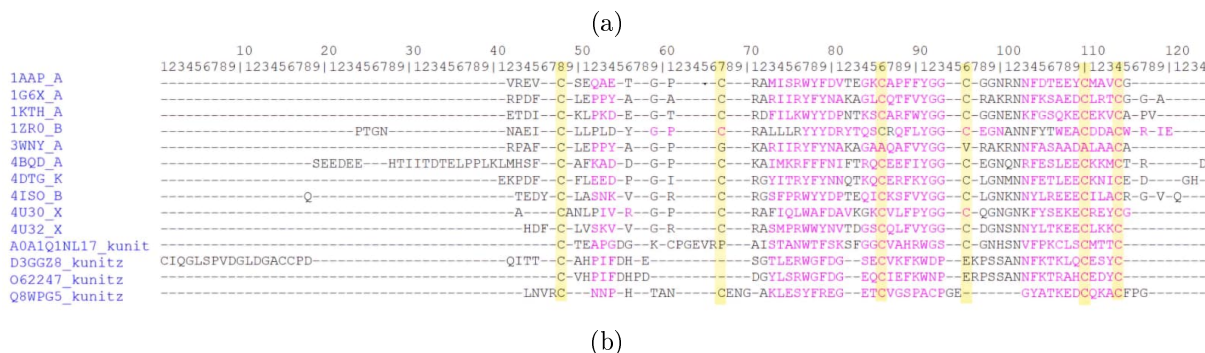


Figure 12: **False negatives analysis based on MSA.** Table in (a) represents the RMSD value obtained from the alignment of the 10 structures used to build the hidden markov model and the 4 false negative investigated in the last part of the project. (b) represent the sequences alignment obtained from the structural alignment, with core region hilighted in purple and the six cysteines hilighted in yellow.

- **Q8WPG5** has higher RMSD values with respect structures used to train the model (2.1-2.9 Å), and it does not have the fourth cysteine (half of the disulfide bridge C2-C4); this high structural divergence with atypical cirteines pattern suggests a very divergent variant of Kunitz domain (potentially a remote homologue).
- **A0A1Q1NL17** RMSD versus training set in average (1.2-1.5 Å) but it does not have the

second cysteine; it can be an atypical Kunitz domain not well represented in the training set.

- **D3GGZ8 and O62247** are the most extreme cases with e-value 0.14 and 0.012; their kunitz domain is positioned away from the N-terminus and, since they are long proteins (195 and 202 aa), this penalizes e-value due to sequence length; they have high RMSD compared to the training set (1.5-1.9 Å) but very low between them (0.353 Å); both of them do not have the second and fourth cysteines (lack of the central C2-C4 disulfide bridge); finally they both belong to nematodes (organisms evolutionarily distant from vertebrates in the training set); these elements together suggest that they belong to an evolutionarily divergent variation of the Kunitz domain, not well represented in the training set built predominantly on vertebrates.